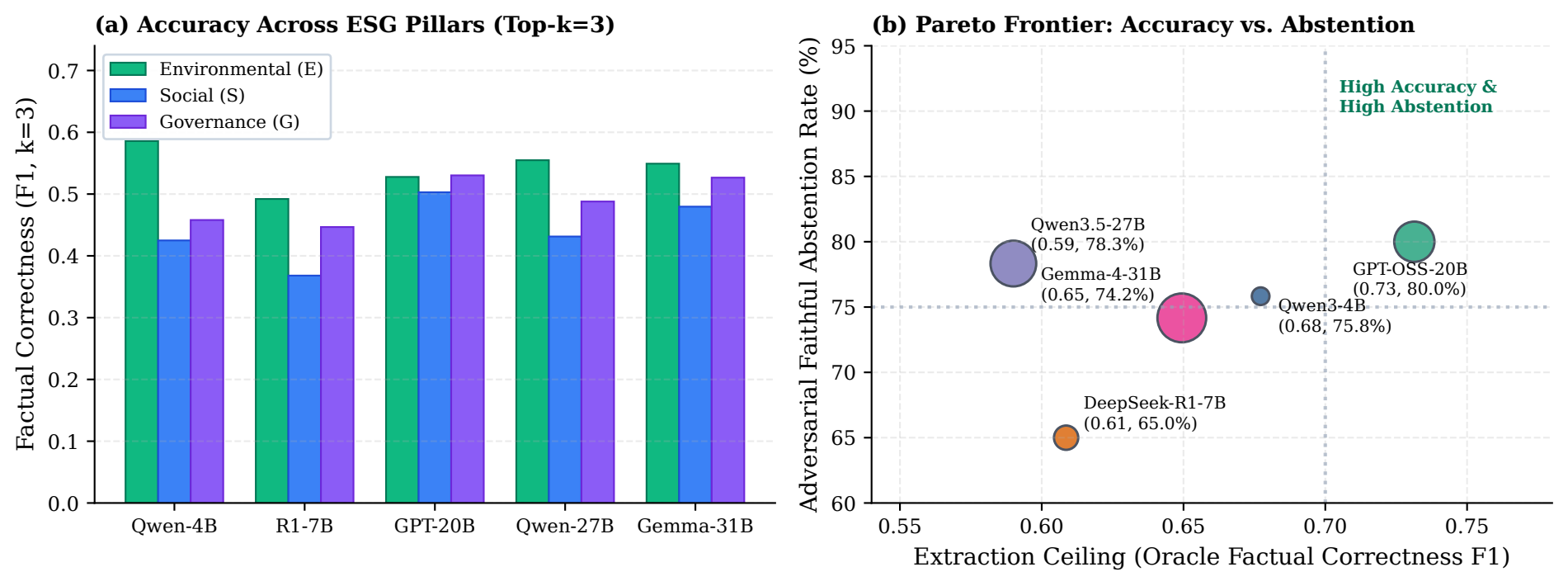
(a) Accuracy Across ESG Pillars (Top-k=3)
Environmental (E)
Social (S)
Governance (G)
Factual Correctness (F1, k=3)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Qwen-4B
R1-7B
GPT-20B
Qwen-27B
Gemma-31B
(b) Pareto Frontier: Accuracy vs. Abstention
High Accuracy &
High Abstention
Qwen3.5-27B
(0.59, 78.3%)
Gemma-4-31B
(0.65, 74.2%)
Qwen3-4B
(0.68, 75.8%)
GPT-OSS-20B
(0.73, 80.0%)
DeepSeek-R1-7B
(0.61, 65.0%)
Adversarial Faithful Abstention Rate (%)
60
65
70
75
80
85
90
95
0.55
0.60
0.65
0.70
0.75
Extraction Ceiling (Oracle Factual Correctness F1)